

6

Some general theorems

In this chapter and the next we shall prove some general results which will find application in Chapter 9. For the present the motivation lies in the evident difficulty of calculating homology groups directly from the definition, and the consequent need for some help from general theorems. Enough is proved here to enable us to calculate the homology groups of all the closed surfaces described in Chapter 2, without difficulty. We shall also calculate the homology groups of some other standard simplicial complexes such as cones (6.8) and spheres (6.11(3)), and show (6.10) that relative homology groups can be defined in terms of ordinary homology groups.

The method presented here for the calculation of homology groups of closed surfaces is based on the idea of collapsing which was introduced in 3.30. This is not the only available method. From the point of view of this book, the advantage of using it is that, having established the invariance of homology groups under barycentric subdivision (5.7), we are in possession of a rapid and rigorous approach to the results. Another method of wide application comes from the Mayer–Vietoris sequence, presented in Chapter 7, where indications are given as to how the homology groups of closed surfaces can be re-calculated by use of the sequence. For effective and unfettered use of the Mayer–Vietoris sequence one really needs to use something like the topological invariance theorem stated in 5.13 – that is, it is better to forget about particular triangulations altogether. Some uninhibited calculations using the Mayer–Vietoris sequence appear in the next chapter, starting at 7.5.

The homology sequence of a pair

Let K be an oriented simplicial complex of dimension n and L a subcomplex of K . Recall from 4.30, 4.32 and 4.34 the homomorphisms $i_{*,j*}$ and ∂_* ; the

following sequence is called the *homology sequence of the pair* (K, L) .

$$0 \rightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{j_*} H_n(K, L) \xrightarrow{\partial_*} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \xrightarrow{j_*} \dots \\ \dots \xrightarrow{j_*} H_1(K, L) \xrightarrow{\partial_*} H_0(L) \xrightarrow{i_*} H_0(K) \xrightarrow{j_*} H_0(K, L) \rightarrow 0$$

Theorem 6.1. *The homology sequence of the pair (K, L) is exact.*

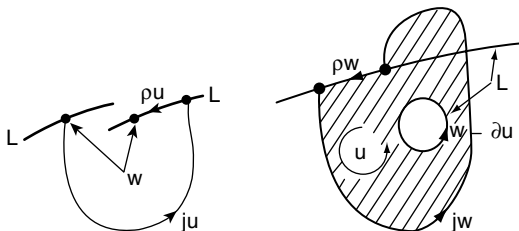
Proof As usual, we regard $C_p(L)$ as a subgroup of $C_p(K)$ (see the note preceding 4.24), and denote by $\rho : C_p(K) \rightarrow C_p(L)$ the ‘restriction’ which puts coefficients of simplexes not in L equal to 0. We have to prove exactness at three places – $H_p(L)$, $H_p(K)$, $H_p(K, L)$ – and each of these involves two proofs of inclusion.

- (1) *Exactness at $H_p(L)$.* Let $z \in Z_{p+1}(K, L)$; then $i_*\partial_*\{z\}_{K,L} = \{\partial z\}_K$. However $z \in C_{p+1}(K)$, so $\partial z \in B_p(K)$, i.e. $\{\partial z\}_K = 0$. Hence $\text{Im } \partial_* \subset \text{Ker } i_*$.

Suppose, conversely, that $w \in Z_p(L)$ and $i_*\{w\}_L = \{w\}_K = 0$. Then $w = \partial u$ for some $u \in C_{p+1}(K)$. Writing $u = ju + \rho u$ we have

$$w = \partial u = \partial ju + \partial \rho u.$$

Hence $\partial ju \in C_p(L)$, that is, $ju \in Z_{p+1}(K, L)$. Since $\partial_*\{ju\}_{K,L} = \{\partial ju\}_L = \{w\}_L - \{\partial \rho u\}_L = \{w\}_L$ ($\partial \rho u \in B_p(L)$), we have $\{w\}_L \in \text{Im } \partial_*$, so $\text{Ker } i_* \subset \text{Im } \partial_*$. Notice that although ∂ju may not equal w (see the left-hand figure), it is homologous to w , since $\partial \rho u \sim 0$ in L .



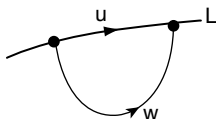
- (2) *Exactness at $H_p(K)$.* Let $z \in Z_p(L)$; then $j_*i_*\{z\}_L = \{jz\}_{K,L} = \{0\} = 0$. Hence $\text{Im } i_* \subset \text{Ker } j_*$. (This is the only one of the three cases where the composite homomorphism is trivial even at the chain level.)

Conversely let $w \in Z_p(K)$ be such that $j_*\{w\}_K = \{jw\}_{K,L} = 0$. Then (see the right-hand figure) $jw = \partial u = j\partial u$ for some $u \in C_{p+1}(K, L)$. Hence $w - \partial u \in C_p(L)$; indeed $\partial(w - \partial u) = \partial w = 0$ so $w - \partial u \in Z_p(L)$.

Finally $i_*\{w - \partial u\}_L = \{w - \partial u\}_K = \{w\}_K$ (since $\partial u \in B_p(K)$), so that $\{w\}_K \in \text{Im } i_*$. This shows $\text{Ker } j_* \subset \text{Im } i_*$.

- (3) *Exactness at $H_p(K, L)$.* Let $z \in Z_p(K)$; then $\partial_* j_* \{z\}_K = \{\partial z\}_L = \{\partial(z - \rho z)\}_L = \{-\partial \rho z\}_L$ ($\partial z = 0$) $= 0$ since $\rho z \in C_p(L)$. (Same diagram as in (2), with z replacing w .) Hence $\text{Im } j_* \subset \text{Ker } \partial_*$.

Conversely, let $w \in Z_p(K, L)$ be such that $\partial_* \{w\}_{K,L} = \{\partial w\}_L = 0$. Thus $\partial w = \partial u$, for some $u \in C_p(L)$, so that $\partial(w - u) = 0 : w - u \in Z_p(K)$. Further $j_* \{w - u\}_K = \{jw - ju\}_{K,L} = \{w\}_{K,L}$. Thus $\{w\}_{K,L} \in \text{Im } j_*$; hence $\text{Ker } \partial_* \subset \text{Im } j_*$.



□

Remark 6.2. There is a slightly different form of the sequence in 6.1 which is sometimes more convenient on account of the fact that it contains reduced zero-dimensional homology groups, and these are zero for connected simplicial complexes (1.23, 4.9(2)). Recall (4.8), $\tilde{H}_0(K) = \text{Ker } \varepsilon_K / \text{Im } \partial_1$, where $\varepsilon_K : C_0(K) \rightarrow \mathbb{Z}$ is the augmentation. For $z \in \text{Ker } \varepsilon_K$ write $[z] = [z]_K = z + \text{Im } \partial_1 \in \tilde{H}_0(K)$.

$$\begin{array}{ll}
 \text{Define } \tilde{\partial}_* & : H_1(K, L) \rightarrow \tilde{H}_0(L) \\
 \text{by} & \{z\}_{K,L} \rightarrow [\partial z]_L \\
 \tilde{i}_* & : \tilde{H}_0(L) \rightarrow \tilde{H}_0(K) \\
 \text{by} & [z]_L \rightarrow [z]_K \\
 \text{and } \tilde{j}_* & : \tilde{H}_0(K) \rightarrow H_0(K, L) \\
 \text{by} & [z]_K \rightarrow \{jz\}_{K,L}.
 \end{array}$$

It is left to the reader to check that these really do define homomorphisms, and that the *reduced sequence of the pair (K, L)* is identical with the above sequence but ending

$$\cdots \rightarrow H_1(K, L) \xrightarrow{\tilde{\partial}_*} \tilde{H}_0(L) \xrightarrow{\tilde{i}_*} \tilde{H}_0(K) \xrightarrow{\tilde{j}_*} \tilde{H}_0(K, L) \rightarrow 0$$

is exact provided L is non-empty.

Examples 6.3. (1) The exact sequence of (K, L) can sometimes be used to derive the relative homology groups of (K, L) from knowledge of the homology groups of L , those of K , and the homomorphisms i_* . The technique is to get

$H_p(K, L)$ in the middle of a short exact sequence:

$$\begin{array}{ccccccc}
 0 & \rightarrow & \text{Ker } \partial_* & \rightarrow & H_p(K, L) & \xrightarrow{\partial_*} & \text{Im } \partial_* \rightarrow 0 \\
 & & \parallel & & & & \parallel \\
 & & \text{Im } j_* & & & & \text{Ker } (i_* : H_{p-1}(L) \rightarrow H_{p-1}(K)) \\
 & & \wr \parallel & & & & \\
 & & H_p(K)/\text{Im}(i_* : H_p(L) \rightarrow H_p(K)) & & & &
 \end{array}$$

If this sequence splits (A.19) then $H_p(K, L)$, including generators, can be calculated – see the examples below.

(2) Let K and L be as in 4.29(3) (Möbius band and its rim). Using the calculations of 4.15, the groups and homomorphisms written in under the following reduced homology sequence are known:

$$\begin{array}{ccccccc}
 H_2(L) & \rightarrow & H_2(K) & \rightarrow & H_2(K, L) & \rightarrow & H_1(L) \xrightarrow{i_*} H_1(K) \xrightarrow{j_*} H_1(K, L) \\
 0 & & 0 & & & & \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \\
 & & & & & & \\
 & & & & \rightarrow \tilde{H}_0(L) & \rightarrow & \tilde{H}_0(K) \rightarrow H_0(K, L) \rightarrow 0 \\
 & & & & 0 & & 0
 \end{array}$$

Hence

$$H_2(K, L) \cong \text{Ker } i_* = 0,$$

$$H_1(K, L) \cong H_1(K)/\text{Im } i_* \cong \mathbb{Z}_2, \text{ with generator the image under } j_* \text{ of a generator of } H_1(K),$$

$$H_0(K, L) = 0.$$

(If we use the unreduced sequence it is only necessary to see that $i_* : H_0(L) \rightarrow H_0(K)$ is an isomorphism (compare 4.31(3)), which implies that the homomorphisms before and after this i_* are trivial.)

(3X) Use the (unreduced) homology sequence to calculate the relative homology groups of (cylinder, both ends) – see 4.29(1) for diagram. In this case $i_* : H_0(L) \rightarrow H_0(K)$ has infinite cyclic kernel, generated by $\{v^1 - v^2\}_L$. Also $i_* : H_1(L) \rightarrow H_1(K)$ takes each generator $\{e^7 + e^8 + e^9\}_L, \{e^{10} + e^{11} + e^{12}\}_L$ to a generator of $H_1(K)$; in particular this i_* is epic (surjective). Use the method of (1) above.

(4X) Take K = the cylinder of 4.29(1) and L = one end. This time the reduced homology sequence of (K, L) (as in (2) above) and the fact that $i_* : H_1(L) \rightarrow H_1(K)$ is an isomorphism show $H_2(K, L)$ and $H_1(K, L)$ are both trivial.

(5X) Show that, for any (K, L) ,

$$H_p(K, L) = 0 \text{ for all } p \Leftrightarrow i_* : H_p(L) \rightarrow H_p(K) \text{ is an isomorphism for all } p.$$

(6) Let K^1 be the 1-skeleton of K ; then $H_1(K^1) = Z_1(K^1) = Z_1(K)$ by 4.9(1) and (3). Similar remarks apply to L and (K, L) ; applying 6.1 to the pair (K^1, L^1) we obtain an exact sequence

$$0 \rightarrow Z_1(L) \rightarrow Z_1(K) \rightarrow Z_1(K, L) \rightarrow H_0(L) \rightarrow H_0(K) \rightarrow H_0(K, L) \rightarrow 0.$$

This was used in the proof of 4.27.

(7X) Let K be a connected oriented graph and L a sub-complex of K with components $L_1, \dots, L_\ell$. Write $\mu(L_\alpha) = \mu_\alpha$ ($\alpha = 1, \dots, \ell$), thus $\mu_\alpha = \text{rank } Z_1(L_\alpha) = \text{rank } H_1(L_\alpha)$; and $\mu(K) = \mu$. Noting that $H_1(K, L)$ is free abelian (4.23(1)), show that $H_1(K, L)$ has rank $\geq \ell - 1$, and that, if the rank is $\ell - 1$ then $\mu = \mu_1 + \dots + \mu_\ell$. (Compare 4.27.) Also, if $\mu = \mu_1 + \dots + \mu_n$, then $i_* : H_1(L) \rightarrow H_1(K)$ is an isomorphism.

(8X) In the example of 4.31(4), calculate $H_1(K, L)$ and $H_2(K, L)$ from the reduced homology sequence of (K, L) . (First verify that $H_2(K) = 0$.)

(9X) Let T be a maximal tree in the 1-skeleton of a connected simplicial complex K . Use the reduced homology sequence of (K, T) to prove that

$$j_* : H_1(K) \rightarrow H_1(K, T)$$

is an isomorphism. Show that the edges of K not in T form a basis for $Z_1(K, T)$. Thus if $e^1, \dots, e^\mu$ are these edges, a presentation of $H_1(K, T)$ is

$$\left(\begin{array}{c} e^1, \dots, e^\mu \\ \widehat{\partial t} = 0 \end{array} \right) \quad \text{for each 2-simplex } t \text{ of } K.$$

Given generators for $H_1(K, T)$ how does one obtain generators for $H_1(K)$? What is the connexion between this method and that of 4.14?

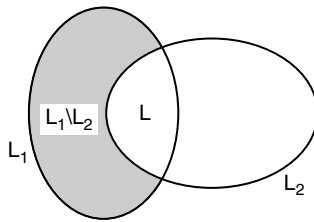
(10X) Use 4.38(4) to show that, in the usual notation,

$$j(Z_p(K)) = Z_p(K, L) \Leftrightarrow i_* : H_{p-1}(L) \rightarrow H_{p-1}(K) \text{ is injective.}$$

Taking $p = 1$, compare with 4.29(4) and 4.31(3).

The excision theorem

The next theorem will be used mainly in Chapter 9, though a few applications will be given here. It derives its name from the fact that it compares the relative homology groups of a pair $(L_1 \cup L_2, L_1)$ with those of a pair $(L_2, L_1 \cap L_2)$ obtained by ‘excising’ $L_1 \setminus L_2$.



Let K be an oriented simplicial complex and L_1, L_2 subcomplexes of K with $K = L_1 \cup L_2$. Write $L = L_1 \cap L_2$. As usual we regard all chain groups as subgroups of those of K .

Theorem 6.4. $H_p(K, L_1) \cong H_p(L_2, L)$ for all p (we could write = here).

Proof The groups $C_p(K, L_1)$ and $C_p(L_2, L)$ are clearly the same subgroup of $C_p(K)$. That is not quite the end of the proof, because we must check that the relative boundary homomorphisms

$$C_p(K, L_1) \rightarrow C_{p-1}(K, L_1) \quad \text{and} \quad C_p(L_2, L) \rightarrow C_{p-1}(L_2, L)$$

coincide. The result will then follow from the definition of relative homology groups. Let

$$j : C_{p-1}(K) \rightarrow C_{p-1}(K, L_1) \quad \text{and} \quad j' : C_{p-1}(L_2) \rightarrow C_{p-1}(L_2, L)$$

be the usual homomorphisms: in each case the homomorphism puts at zero the coefficient of any simplex not in L_1 , so that if $c \in C_{p-1}(L_2)$ then $jc = j'c \in C_{p-1}(K, L_1) = C_{p-1}(L_2, L)$. Hence, for any $c' \in C_p(K, L_1) = C_p(L_2, L)$, we have $j\partial c' = j'\partial c'$. This is the end of the proof. $\square$

Example 6.5. Let M be a two-dimensional simplicial complex and s a 2-simplex of M . Then removing s from M leaves a subcomplex K say. We ask the following question: when does $H_1(K)$ depend, up to isomorphism, only on M and not on the particular 2-simplex s which was removed? Now $M = K \cup \bar{s}$, $\bar{s} = K \cap \bar{s}$, where, as usual, $\bar{s}$ and $\dot{s}$ are the set of faces and proper faces of s , respectively. By excision $H_p(M, K) = H_p(\bar{s}, \dot{s})$, which is isomorphic to $\mathbb{Z}$ if $p = 2$ and to 0 if $p = 1$, by a straightforward calculation (or see 4.23(4)).

Consider the exact sequence (6.1)

$$H_2(K) \rightarrow H_2(M) \rightarrow H_2(M, K) \rightarrow H_1(K) \rightarrow H_1(M) \rightarrow H_1(M, K)$$

$$\mathbb{Z} \qquad \qquad \qquad 0$$

If $H_1(M)$ is free abelian and $H_2(M) = 0$ then it follows from the exact sequence that $H_1(K) \cong H_1(M) \oplus \mathbb{Z}$ and therefore that $H_1(K)$ does not depend on the choice of s . These conditions are not necessary, however, for if M is a closed surface the same results holds (this is proved in the course of proving 6.13). (XX) The reader is invited to find an M for which $H_1(K)$ does depend on s and to explain this dependence from the exact sequence.

Collapsing revisited

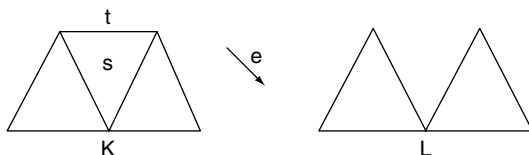
The idea of collapsing was introduced in 3.30, but the reason for introducing it is the following result.

Theorem 6.6. *Suppose that an oriented simplicial complex K collapses to a subcomplex L . Then the homomorphism*

$$i_* : H_p(L) \rightarrow H_p(K)$$

is an isomorphism for all p .

Proof Suppose first that K collapses to L by an elementary collapse, so that $K = L \cup \{s, t\}$ say where s is a principal q -simplex of K and t is a free face of s .



We shall prove that $H_p(K, L) = 0$ for all p ; the result then follows immediately from the homology sequence of (K, L) .

The chain complex of (K, L) contains only two non-zero relative chain groups, namely those in dimensions q and $q - 1$, and therefore takes the form

$$\cdots \rightarrow 0 \rightarrow C_q(K, L) \xrightarrow{\hat{\partial}} C_{q-1}(K, L) \rightarrow 0 \rightarrow \cdots$$

Now $C_q(K, L)$ is infinite cyclic, generated by the oriented simplex σ corresponding to s , and $C_{q-1}(K, L)$ is also infinite cyclic, generated by the oriented simplex τ corresponding to t . Furthermore $\hat{\partial}\sigma = \pm\tau$ so that $\hat{\partial}$ is an isomorphism. It follows from the definition of relative homology groups that $H_p(K, L) = 0$ for all p .

For the general case, consider a sequence of elementary collapses $K \searrow^e K_1 \searrow^e K_2 \searrow^e \cdots \searrow^e K_m = L$. It is straightforward to verify that the homomorphism

$$i_* : H_p(L) \rightarrow H_p(K)$$

is the composite

$$H_p(K_m) \rightarrow H_p(K_{m-1}) \rightarrow \cdots \rightarrow H_p(K_1) \rightarrow H_p(K)$$

where each homomorphism is given by the inclusion of a subcomplex. Consequently i_* is an isomorphism by the first part. $\square$

Note that this theorem does not just tell us $H_p(K)$ up to isomorphism: it says that p -cycles whose homology classes on L generate $H_p(L)$ have homology classes on K which generate $H_p(K)$.

Examples 6.7. (1) The results in 4.15 and 4.17(1) (the homology groups of Möbius band and cylinder) become obvious, using 6.6, since each simplicial complex collapses to a simple closed polygon.

(2) As a special case of 6.6, if K collapses to a point, then $H_p(K) = 0$ ($p > 0$); $H_0(K) \cong \mathbb{Z}$. If K is a graph, then the converse is true – see 1.12(2). However there exist two-dimensional simplicial complexes which satisfy the homological condition just given but which cannot be collapsed at all since there is no principal simplex with a free face. An example is given by any triangulation of the ‘dunce hat’; see 3.22(4).

Before going on to calculate the homology groups of closed surfaces by collapsing, here is a simple application to cones.

Let K be a simplicial complex in $\mathbb{R}^N$ and let $v \in \mathbb{R}^N$ be a vector not affinely dependent on the vertices of K (if necessary add one to N in order to find such a v). The *cone on K with vertex v* , denoted CK , is the simplicial complex whose simplexes are of the form

$$(v) \quad \text{or} \quad (vv^0 \dots v^p) \quad \text{or} \quad (v^0 \dots v^p)$$

where $(v^0 \dots v^p) \in K$. (K is called the *base* of the cone.) Thus CK is the join of K to the simplicial complex $\{v\}$ (see 3.26), and, up to isomorphism, is independent of the choice of v . If K is oriented then CK can be oriented by regarding the above elements of CK as oriented simplexes.

Theorem 6.8. (Homology groups of a cone) *With the above notation,*

$$H_p(CK) = 0 \quad \text{for } p > 0$$

while $H_0(CK) \cong \mathbb{Z}.$

Proof We prove CK collapses to a point – in fact to v – and the result then follows from 6.6. If K is empty then $CK = \{v\}$, so there is nothing to collapse (or to prove). Otherwise we proceed by induction on the dimension of K . If $\dim K = n$, then any $(n + 1)$ -simplexes of CK can be collapsed, removing the n -simplexes of K and therefore leaving a cone on a simplicial complex of dimension $n - 1$. If $\dim K = 0$ the same procedure collapses CK directly to the vertex of the cone. $\square$

Corollaries 6.9. (1) *Suppose that L is a subcomplex of K and that L is a cone. Then*

$$j_* : H_p(K) \rightarrow H_p(K, L) \quad p > 0$$

and $\tilde{j}_* : \tilde{H}_0(K) \rightarrow H_0(K, L) \quad (\text{see 6.2})$

are isomorphisms.

(2) *For any K ,*

$$H_p(CK, K) \cong H_{p-1}(K) \quad p > 1$$

$$H_1(CK, K) \cong \tilde{H}_0(K)$$

(3) (Homology groups of a simplex) *Let s be an n -simplex ($n \geq 0$) and $K = \bar{s}$ (oriented arbitrarily). Then $H_p(K) = 0$ for $p > 0$ and $H_0(K) \cong \mathbb{Z}$.*

Proof (1) The homology sequence of the pair (K, L) contains the section

$$H_p(L) \rightarrow H_p(K) \xrightarrow{j_*} H_p(K, L) \rightarrow H_{p-1}(L).$$

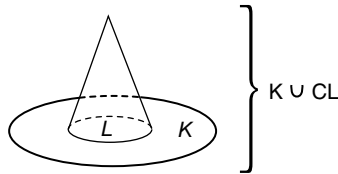
For $p > 1$, both end-groups are zero, by 6.8, and the result follows from exactness. For $p = 0$, $p = 1$ the result follows from a similar application of the reduced homology sequence.

(2) The proof is similar to that of (1).

(3) K is in fact a cone, with empty base if $n = 0$ and with base the closure of an $(n - 1)$ -dimensional face of s if $n > 0$. Thus the result follows from 6.8. $\square$

Using the excision theorem and 6.8 we can establish the following interpretation of relative homology groups as the absolute homology groups of an

associated simplicial complex. The idea is that the relative homology groups of the pair (K, L) in some sense disregard L , and homologically this can be done by adding a cone on L . Thus let $K \cup CL$ be obtained by taking the union of K and a cone CL on L , the cone meeting K precisely on L as in the sketch.



Theorem 6.10. *With the above notation*

$$H_p(K, L) \cong H_p(K \cup CL) \quad p > 0$$

and

$$H_0(K, L) \cong \tilde{H}_0(K \cup CL).$$

(N.B. If L is empty, then CL is a single point not in $|K|$.)

Proof By the excision theorem 6.4,

$$H_p(K \cup CL, CL) = H_p(K, K \cap CL) = H_p(K, L)$$

for all p . The result follows from 6.9(1). □

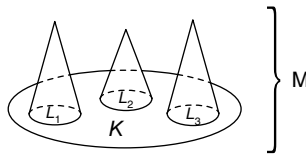
Examples 6.11. (1) Let K be the Möbius band given in 4.15 and let L be the rim of the band. Adding a cone on L gives precisely the projective plane in 4.17(5). The relative homology groups of (K, L) are given in 4.29(3). (It is difficult to visualize $K \cup CL$ in this case because a Möbius band cannot be realized in $\mathbb{R}^3$ in such a way that the circular rim is a plane polygon with the band all on one side of it. With a possibly larger number of triangles in the Möbius band we can do the construction in $\mathbb{R}^4$; compare 2.16(5).)

(2X) Let K be connected, and let the subcomplex L have components $L_1, \dots, L_\ell$. Then we can form a simplicial complex M by adding disjoint cones on the subcomplexes $L_1, \dots, L_\ell$ as in the diagram below. This differs from $K \cup CL$ in that the vertices of the cones are all distinct.

$$\text{Show that} \quad H_p(K, L) \cong H_p(M) \quad p > 1$$

and

$$H_1(K, L) \cong H_1(M) \oplus \mathbb{Z}^{\ell-1}.$$

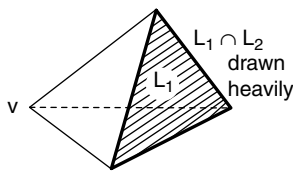


As an example of this let K be the cylinder given in 4.29(1) and L_1 and L_2 the two ends. Then M is a closed surface which is in fact a sphere, i.e. has standard form aa^{-1} (2.8). The relative homology groups of $(K, L_1 \cup L_2)$ are in 4.29(1) and the homology groups of any closed surface are given in 4.18 and 6.13 below.

(3X) Suppose that K, L and M are simplicial complexes of dimension 0, each consisting of $n \geq 1$ vertices. Suppose that K and L are joinable, and also that KL and M are joinable (3.26); denote the join of KL and M by KLM . (Thus when $n = 2$ the join KLM is an octahedron.) Show that $H_1(KLM) = 0$. [Hint. Take a maximal tree in the graph KL and extend it to a maximal tree in the 1-skeleton of KLM . Show, using 6.8, that the basic 1-cycles (4.9(2)) are homologous to zero.]

(4) Let S^n denote the boundary (i.e. the set of proper faces) of an $(n + 1)$ -simplex σ_{n+1} . Thus S^0 is two points, S^1 is a ‘hollow’ triangle, S^2 is a hollow tetrahedron. (In general, S^n is homeomorphic to $\{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_1^2 + \dots + x_{n+1}^2 = 1\}$ and can be called an n -sphere.) The homology groups of S^0 present no trouble, so suppose $n > 0$ and consider the following two subcomplexes of S^n . Let v be a vertex of σ_{n+1} and let σ_n be the n -simplex whose vertices are the remaining $n + 1$ vertices of σ_{n+1} .

Define L_1 = the set of all faces of σ_n .
 L_2 = the cone with vertex v on the set of all proper faces of σ_n .
(The diagram shows $n = 2$.)



a closed surface has no principal simplex with a free face. However we can remove a 2-simplex and, using 3.34, collapse what is left to a graph, whose first homology group is straightforward to calculate. We need to know the effect of removing a 2-simplex, and that is the content of the next Lemma.

Let M be a closed surface, and let K be obtained from M by removing a single 2-simplex s ; thus K is a subcomplex of M . Let z be the 1-cycle on K going once round the boundary of s and let A be the subgroup of $H_1(K)$ generated by $\{z\}_K$. (Of course $\{z\}_M = 0$ since z bounds s .)

Lemma 6.12. (1) *Suppose that M is orientable. Then*

$$i_* : H_1(K) \rightarrow H_1(M)$$

is an isomorphism.

(2) *Suppose that M is non-orientable. Then there is a short exact sequence*

$$0 \rightarrow A \xrightarrow{\text{inclusion}} H_1(K) \xrightarrow{i_*} H_1(M) \rightarrow 0.$$

It follows that $H_1(M) \cong H_1(K)/A$ and generators for $H_1(M)$ can be obtained by taking the images under i_ of generators for $H_1(K)$.*

Proof Consider the homology sequence of the pair (M, K) :

$$\begin{array}{ccccccc} H_2(M) & \xrightarrow{j_*} & H_2(M, K) & \xrightarrow{\partial_*} & H_1(K) & \xrightarrow{i_*} & H_1(M) \rightarrow H_1(M, K) \\ \parallel & & & & \parallel & & \\ H_2(\bar{s}, \dot{s}) & \cong & \mathbb{Z} & & H_1(\bar{s}, \dot{s}) & = & 0 \end{array}$$

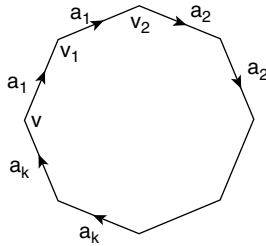
Here $\bar{s}$ and $\dot{s}$ are respectively the set of faces of s and the set of proper faces – the boundary – of s . The equalities follow by excision, and by 4.23(4), which gives the homology groups of $(\bar{s}, \dot{s})$.

(1) Suppose that M is orientable. Let σ be the oriented 2-simplex of M corresponding to s . Now M has a coherent orientation in which s receives the orientation of σ ; also $H_2(M) \cong \mathbb{Z}$, with generator the sum of all coherently oriented 2-simplexes of M (4.18). Since j_* suppresses all 2-simplexes of M but σ , it follows that j_* will take the generator of $H_2(M)$ to $\{\sigma\}$, the homology class of σ in $H_2(M, K) = H_2(\bar{s}, \dot{s})$. However $\{\sigma\}$ generates $H_2(\bar{s}, \dot{s})$ so j_* is an isomorphism. Hence $\partial_* = 0$ and i_* is an isomorphism by exactness.

(2) Suppose that M is non-orientable, Then $H_2(M) = 0$ (4.18). As in (1), $H_2(M, K) \cong \mathbb{Z}$, with generator $\{\sigma\}$. Now $\partial_*\{\sigma\} = \{\partial\sigma\}_K = \{\pm z\}$, i.e. $\text{Im } \partial_* = A$. This proves the result, since $\text{Im } \partial_* = \text{Ker } i_*$. $\square$

Now consider a closed surface M represented in the standard form of 2.8. According to 3.34, removing a 2-simplex produces a simplicial complex K which collapses to the graph G in M given by the boundary of the polygonal region. Note that this collapse really does occur in K since the only identifications to be made in passing from the polygonal representation to the closed surface are identifications round the boundary polygon, and this is not touched during the collapse. It would not be valid, for instance, to invoke 3.32 to prove that M collapses to a point, since the collapsing of a disk starts from the boundary polygon, and the edges of M around the polygon are not in reality free.

We must now identify the graph G . Let us consider the case 2.8(2) for definiteness. Then the symbol is $a_1 a_1 \dots a_k a_k$, as in the diagram.



Each a_i will contain several vertices and edges of M , but the $2k$ vertices of the polygon at the endpoints of the a_i are all the same vertex v of M . (For $v_1 = v$ because they are both at the beginning of a_1 , $v_2 = v_1$ because they are both at the end of a_1 , and so on.) Thus each a_i gives a loop on M and all these loops have the vertex v of M in common. The case 2.8(3) is similar, and 2.8(1), the sphere aa^{-1} , differs only in that the end-points of a are distinct vertices of M . Thus the graph G to which K collapses is as follows:

- (1) a simple polygonal arc in the case of the sphere aa^{-1} ;
- (2) k loops, given by $a_1, \dots, a_k$, with a single vertex in common, in the case of the non-orientable closed surface $a_1 a_1 \dots a_k a_k$;
- (3) $2h$ loops, given by $a_1, b_1, \dots, a_h, b_h$, with a single vertex in common, in the case of the orientable closed surface $a_1 b_1 a_1^{-1} b_1^{-1} \dots a_h b_h a_h^{-1} b_h^{-1}$. (See 3.35(1) for the case $h = 1$.)

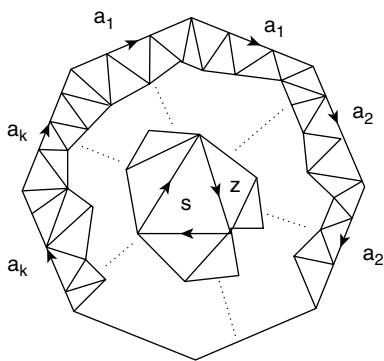
Since K collapses to G , the map $i_* : H_1(G) \rightarrow H_1(K)$ is an isomorphism by 6.6. The group $H_1(G)$ is isomorphic to $\mathbb{Z}^{\mu(G)}$ and has generators given by the

basic loops on G (see 4.9(3)). Thus

$$H_1(K) \cong \begin{cases} 0 & \text{in Case 1} \\ \mathbb{Z}^k, & \text{and is generated by } \{a_1\}, \dots, \{a_k\} \text{ in Case 2} \\ \mathbb{Z}^{2h}, & \text{and is generated by } \{a_1\}, \{b_1\}, \dots, \{a_h\}, \{b_h\}, \text{ in Case 3} \end{cases}$$

When M is non-orientable (Case 2), we need to know the homology class on K represented by the 1-cycle going round the boundary of a triangle s in M . Then we can apply 6.12(2). The inside of the polygon is broken up into triangles, and by adding up the boundaries of all the triangles except s , oriented clockwise in the picture, we have

$$a_1 + a_1 + a_2 + a_2 + \dots + a_k + a_k \sim z \text{ in } K.$$



Thus the group A of 6.12(2) is generated by $2\{a_1\} + \dots + 2\{a_k\} \in H_1(K)$. (There is little difficulty proving that the triangles of a triangulation of a disk can be oriented coherently in this way: an induction similar to that in 3.32 will work. Of course this does not coherently orient M – an impossibility – since the orientations are not coherent across the edges of M making up any of the a_i .)

Theorem 6.13. *Let M be a closed surface. Then $H_1(M)$ is as follows.*

- (1) $H_1(M) = 0$ when M has standard form aa^{-1} .
- (2) $H_1(M) \cong \mathbb{Z}^{k-1} \oplus \mathbb{Z}_2$ for non-orientable closed surface $a_1a_1 \dots a_k a_k$. The classes $\{a_1\}, \dots, \{a_k\}$ on M generate $H_1(M)$ and the unique element of order two is $\{a_1\} + \dots + \{a_k\}$.
- (3) $H_1(M) \cong \mathbb{Z}^{2h}$ for the orientable closed surface $a_1b_1a_1^{-1}b_1^{-1} \dots a_hb_ha_h^{-1}b_h^{-1}$. The classes $\{a_1\}, \{b_1\}, \dots, \{a_h\}, \{b_h\}$ freely generate $H_1(M)$.

Proof The cases (1) and (3) follow at once from 6.12(1) and the group $H_1(K)$, given above. Now for the case (2). The exact sequence of 6.12(2) shows that

$H_1(M)$ is generated by the images under i_* of the generators of $H_1(K)$, i.e. by $i_*\{a_j\}_K = \{a_j\}_M$ for $j = 1, \dots, k$. Write $\alpha_j = \{a_j\}_K$. Now $H_1(M) \cong H_1(K)/A$ and this is the quotient group of the free abelian group generated by $\alpha_1, \dots, \alpha_k$, by the subgroup generated by 2α , where $\alpha = \alpha_1 + \dots + \alpha_k$. Changing generators for $H_1(K)$ to $\alpha_1, \dots, \alpha_{k-1}, \alpha$, the quotient has presentation

$$\left(\begin{array}{l} \alpha_1, \dots, \alpha_{k-1}, \alpha \\ 2\alpha = 0 \end{array} \right)$$

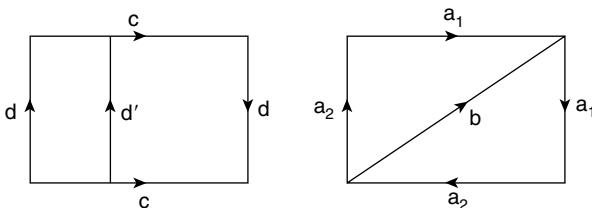
and is therefore isomorphic to $\mathbb{Z}^{k-1} \oplus \mathbb{Z}_2$. The unique element of order two is the coset $\alpha + A$ corresponding to α , so the unique element of order two in $H_1(M)$ is

$$i_*\alpha = \{a_1 + \dots + a_k\}_M = \{a_1\}_M + \dots + \{a_k\}_M.$$

□

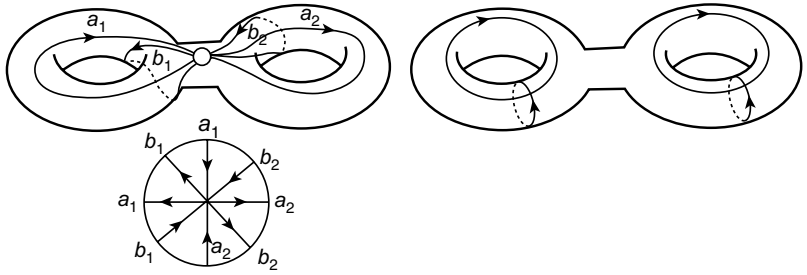
Thus two closed surfaces M_1 and M_2 have the same standard form (i.e. are equivalent) if and only if $H_1(M_1) \cong H_1(M_2)$: the first homology group is a sufficiently delicate invariant to distinguish the closed surfaces from one another. (In contrast, it does not distinguish a cylinder from a Möbius band, nor either of these from a simple closed polygon!)

Examples 6.14. (1X) If M is a sphere with one crosscap (projective plane) then $H_1(M) \cong \mathbb{Z}_2$. This agrees with 4.17(5). For M a sphere with two crosscaps (Klein bottle), $H_1(M) \cong \mathbb{Z} \oplus \mathbb{Z}_2$. The usual representation of a Klein bottle is shown below on the left (compare 2.2(4)) and the standard one on the right. Cutting along the diagonal b in the right-hand diagram and sticking together the two a_1 's produces the left-hand picture with $c = a_2$ and $d = b$. Also $b \sim a_1 + a_2$ (add up the boundaries of triangles in the 'top half' of the right-hand picture), so by 6.13(2) $H_1(M)$ is generated by $\{c\}$ and $\{d\}$, and $\{d\}$ is the unique element of order two. To see that $d + d \sim 0$, note that $d \sim d'$ from the left half of the left-hand picture and $d' \sim -d$ from the right half. (Compare 4.29(3).)

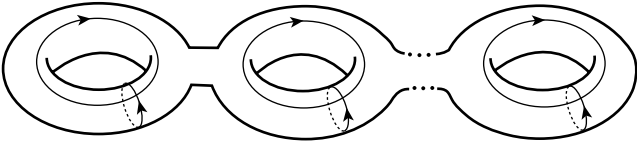


Let z be the 1-cycle going across the left-hand diagram between the mid-points of d , parallel to c . What is $\{z\}$ in terms of $\{c\}$ and $\{d\}$? What is $2\{z\}$? (In particular $2\{z\} \neq 0$.)

(2) Taking $h = 2$ in 6.13(3), possible positions for a_1, b_1, a_2, b_2 on a double torus are shown in the diagram on the left. These 1-cycles are homologous to the four shown in the diagram on the right. These four accordingly represent free generators for $H_1(M)$.



Similarly the 1-cycles drawn below, for arbitrary h , are homologous to $a_1, b_1, \dots, a_h, b_h$ and so represent free generators for $H_1(M)$.



(3X) The method used to prove 6.13 can of course be used to calculate the first homology groups of a closed surface given in any polygonal representation, not just the standard one. The reader may care to try the examples of 2.16(1)–(4) directly from the representations given there.

The Euler characteristic

In this section we shall establish a connexion between the Euler characteristic of a simplicial complex (i.e. the alternating sum of the numbers of p -simplexes) and the homology groups of the simplicial complex. In particular this implies using the invariance theorem 5.13, that the Euler characteristic is a topological invariant of a simplicial complex: it is unaffected by homeomorphisms. It is interesting to note that though the Euler characteristic can be defined without mentioning homology groups, the only way (so far as I know) of proving its

topological invariance is by the use of homology theory. The invariance theorem says that two simplicial complexes with different Euler characteristics cannot have homeomorphic underlying spaces. The converse is not true, even for closed surfaces, for hT and $2hP$ have the same characteristic (namely $2 - 2h$) for any $h > 0$.

Definition 6.15. Let K be a simplicial complex of dimension n and L a subcomplex of K . Let $\alpha_p(K)$ and $\alpha_p(K, L)$ denote the numbers of p -simplexes in K and in $K \setminus L$ respectively. Then

$$\chi(K) = \sum_{p=0}^n (-1)^p \alpha_p(K) \quad \text{and} \quad \chi(K, L) = \sum_{p=0}^n (-1)^p \alpha_p(K, L)$$

are called the *Euler (or Euler–Poincaré) characteristics* of K and of $K \bmod L$ respectively. (Compare 2.11.)

The connexion between these characteristics and the homology groups of K and of (K, L) is given in the next theorem, in which the K of 6.15 is assumed oriented. Let

$$\beta_p(K) = \text{rank } H_p(K) \quad \text{and} \quad \beta_p(K, L) = \text{rank } H_p(K, L).$$

(See A.29 for the definition of rank of a finitely generated abelian group.) These numbers are called the *p th Betti numbers* of K and of $K \bmod L$ respectively. (Betti numbers were so called by H. Poincaré in honour of the Italian mathematician Enrico Betti.)

Theorem 6.16.

$$\chi(K) = \sum_{p=0}^n (-1)^p \beta_p(K); \quad \chi(K, L) = \sum_{p=0}^n (-1)^p \beta_p(K, L).$$

Proof The result is a purely algebraic one. Given any sequence of finitely generated abelian groups

$$0 \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots \rightarrow C_p \xrightarrow{d_p} \cdots \rightarrow C_1 \xrightarrow{d_1} C_0 \xrightarrow{d_0} 0$$

where $d_{p-1} \circ d_p = 0$ for $p = 1, \dots, n$, we can define $H_p = \text{Ker } d_p / \text{Im } d_{p+1}$ ($p = 0, \dots, n$). (Compare 4.36.) Writing $\alpha_p = \text{rank } C_p$, $\beta_p = \text{rank } H_p$, we shall prove $\sum_{p=0}^n (-1)^p \alpha_p = \sum_{p=0}^n (-1)^p \beta_p$. The results required are then immediate.

We have short exact sequences

$$\begin{aligned} 0 &\rightarrow \text{Ker } d_p \rightarrow C_p \xrightarrow{d_p} \text{Im } d_p \rightarrow 0 \\ \text{and} \quad 0 &\rightarrow \text{Im } d_{p+1} \rightarrow \text{Ker } d_p \rightarrow H_p \rightarrow 0. \\ \text{Thus by A.31,} \quad \alpha_p &= \text{rank Ker } d_p + \text{rank Im } d_p, \\ \text{and} \quad \beta_p &= \text{rank Ker } d_p - \text{rank Im } d_{p+1} \end{aligned}$$

so that $\alpha_p - \beta_p = \text{rank Im } d_p + \text{rank Im } d_{p+1}$ and $\sum_{p=0}^n (-1)^p (\alpha_p - \beta_p) = 0$ since d_0 and d_{n+1} are both trivial. $\square$

Notice that A.31(3) is the special case of this result which arises when the long sequence is exact, in which case $H_p = 0$ for all p .

Examples 6.17. (1) From the definitions of chain groups it follows that, in the notation of 6.15, $\chi(K) = \chi(L) + \chi(K, L)$. This says something not quite so obvious about homology groups in view of 6.16.

Similarly if L_1 and L_2 are subcomplexes of a simplicial complex K then clearly

$$\chi(L_1 \cup L_2) = \chi(L_1) + \chi(L_2) - \chi(L_1 \cap L_2).$$

(2X) Using 6.16 and the calculated homology groups of closed surfaces in 4.18 and 6.13 we can check the values of their Euler characteristics given in 2.14.

(3X) In the notation of 6.11(3), we can calculate the Euler characteristic of KLM directly. In fact (compare 3.27(5))

$$\begin{aligned} \alpha_0(KL) &= 2n; \quad \alpha_1(KL) = n^2 \\ \alpha_0(KLM) &= 3n; \quad \alpha_1(KLM) = n^2 + 2n^2 = 3n^2; \quad \alpha_2(KLM) = n^3. \end{aligned}$$

Hence $\chi(KLM) = 3n - 3n^2 + n^3$. Since $H_0(KLM) \cong \mathbb{Z}$ and $H_1(KLM) = 0$, it follows from 6.16 that $\text{rank } H_2(KLM) = (n-1)^3$. However $H_2(KLM)$ is free abelian since KLM is two-dimensional (compare 4.9(3)), so that $H_2(KLM) \cong \mathbb{Z}^{(n-1)^3}$.

(4X) Let s be an n -simplex and let $K = \bar{s}$. Denote as usual by K^q the q -skeleton of K . Now from 6.9(3), $H_p(K) = 0$ if $0 < p < n$; consequently

(4.9(1)) $H_p(K^q) = 0$ if $0 < p < q \leq n$. Also $\alpha_p(K) = \binom{n+1}{p+1}$ since every choice of $p+1$ vertices from among the vertices of s spans a p -simplex of K . In an analogous way to (3) above, show that $H_q(K^q)$ is a free abelian group of rank $\binom{n}{q+1}$ if $0 < q < n$. In particular putting $q = n-1$ we get an alternative proof of 6.11(4): $H_{n-1}(S^{n-1}) \cong \mathbb{Z}$ for all $n > 1$.